

Forms of Quadratic Functions	$y = ax^2 + bx + c$	$y = a(x - h)^2 + k$	$y = a(x - p)(x - q)$
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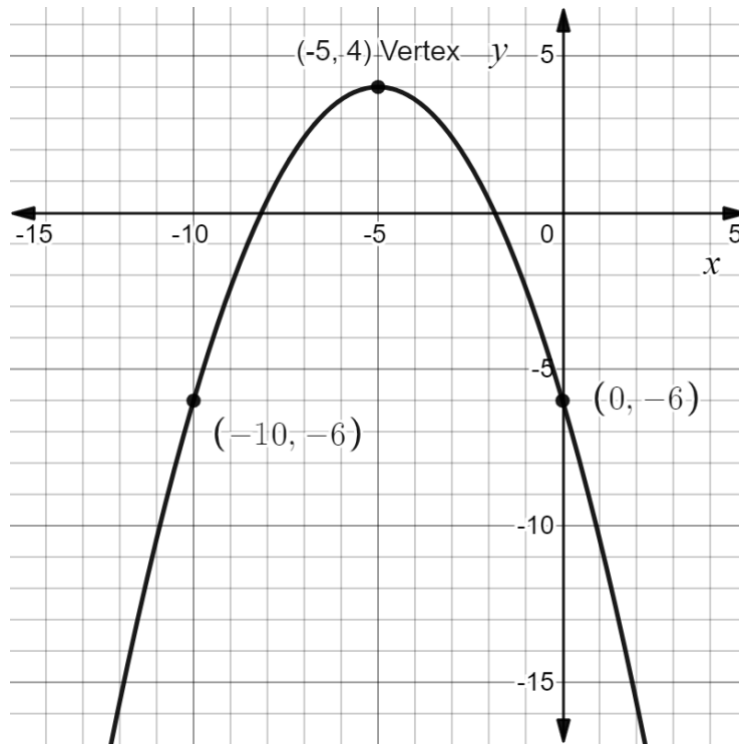
1. Write the equation for a quadratic function that has x -intercepts $(-2, 0)$ and $(7, 0)$ and passes through the point $(5, -42)$.
 - a. Write the equation for the quadratic function in factored form. Show your work in the space provided.

Work	
Factored Form	

- b. Rewrite the equation for the quadratic function that has x -intercepts $(-2, 0)$ and $(7, 0)$ and passes through the point $(5, -42)$ in standard form. Show your work in the space provided.

Work	
Standard Form	

2. A quadratic function is graphed on the coordinate grid. The vertex and two points equidistant from the vertex are labeled.



Write the equation for the quadratic function graphed in vertex form.

Work	
Vertex Form	

3. A table of values for a quadratic function is shown.

x	-3	1	3	5	9
y	-44	84	100	84	-44

- a. Write the equation in vertex form for the quadratic function using the table.

Work	
Vertex Form	

- b. Rewrite the equation for the quadratic function in the table in standard form. Show your work in the space provided.

Work	
Standard Form	

4. The US government monitors the consumption of different products. The table shows y , the amount of ice cream consumed in millions of pounds, for x years since 2010.

x	0	1	2	3	4	5	6	7	8
y	4,354	4,222	4,114	4,030	3,970	3,934	3,922	3,934	3,970

Write the quadratic equation, in any form, for the quadratic function that models the amount of ice cream consumed, in millions of pounds, since 2010.

Work	
Equation	